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Biological Oxidation (Electron transport chain – Chemiosmosis – Oxidative Phosphorylation – Uncouplers)

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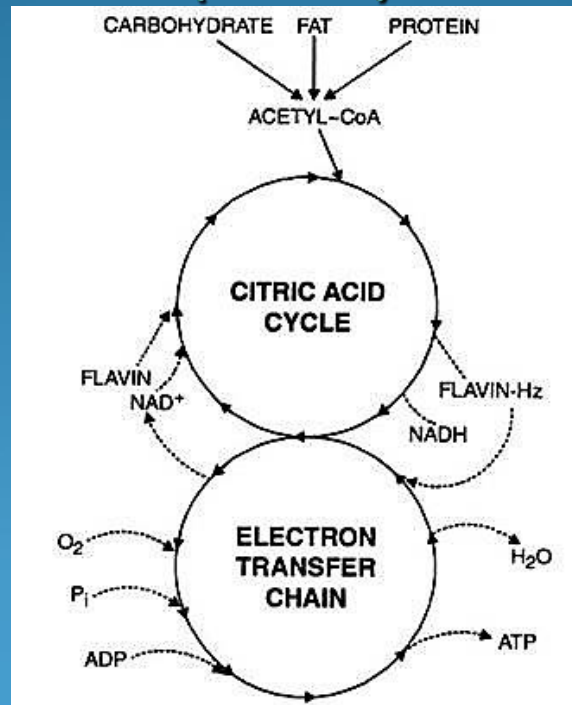
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Biological Oxidation

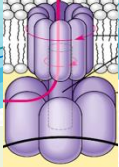
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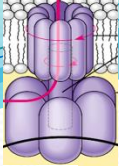
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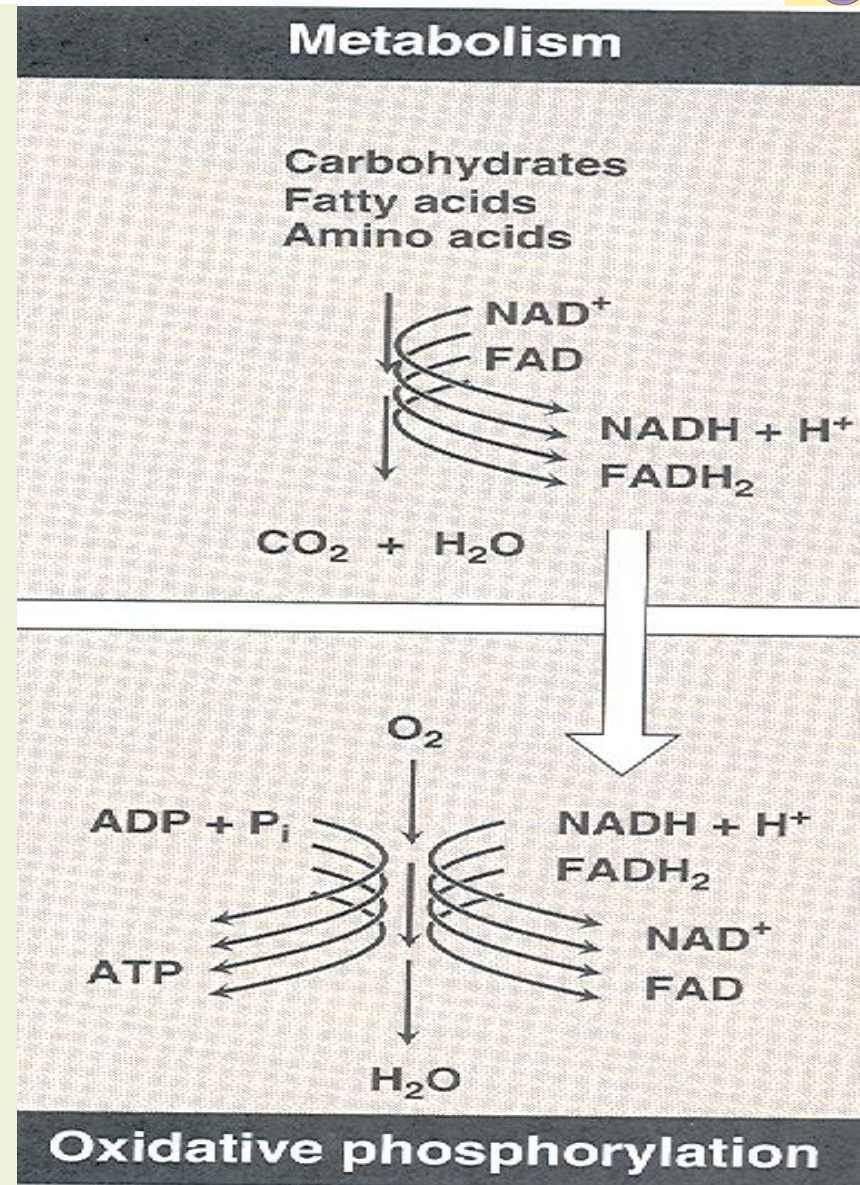
- **Bioenergetics:** It describes the transfer and utilization of energy in biological system.
- All the catabolic products of the food components → (CHO, fats and proteins) are **metabolized** into principle sources of **reducing equivalents** (such as **NAD** & **FAD**). These NAD and FAD have a high **transfer** [redox] potentials.



- **Electron Transport:** Electrons carried by reduced coenzymes (**NADH** & **FADH**) are passed through a chain of proteins and coenzymes to drive the generation of a **electrochemical or proton gradient across** the inner mitochondrial membrane.



- **Redox potential** → Electron affinity
- **Oxygen** has the **highest** electron affinity (↑↑↑ highest-redox-potential), **electrophilic**. Hydrogen has the **lowest** electron affinity (↓↓↓ lowest redox potential), **nucleophilic**.
- **Oxidative phosphorylation** is the process of converting this high redox potential into energy-rich ATP molecules.
- **$RH_2 \rightarrow NADH$ & $FADH \rightarrow ETC$**
Proteins (electron transport + H^+ pumping → Electrochemical Gradient) → $O_2 \rightarrow H_2O + ATP$



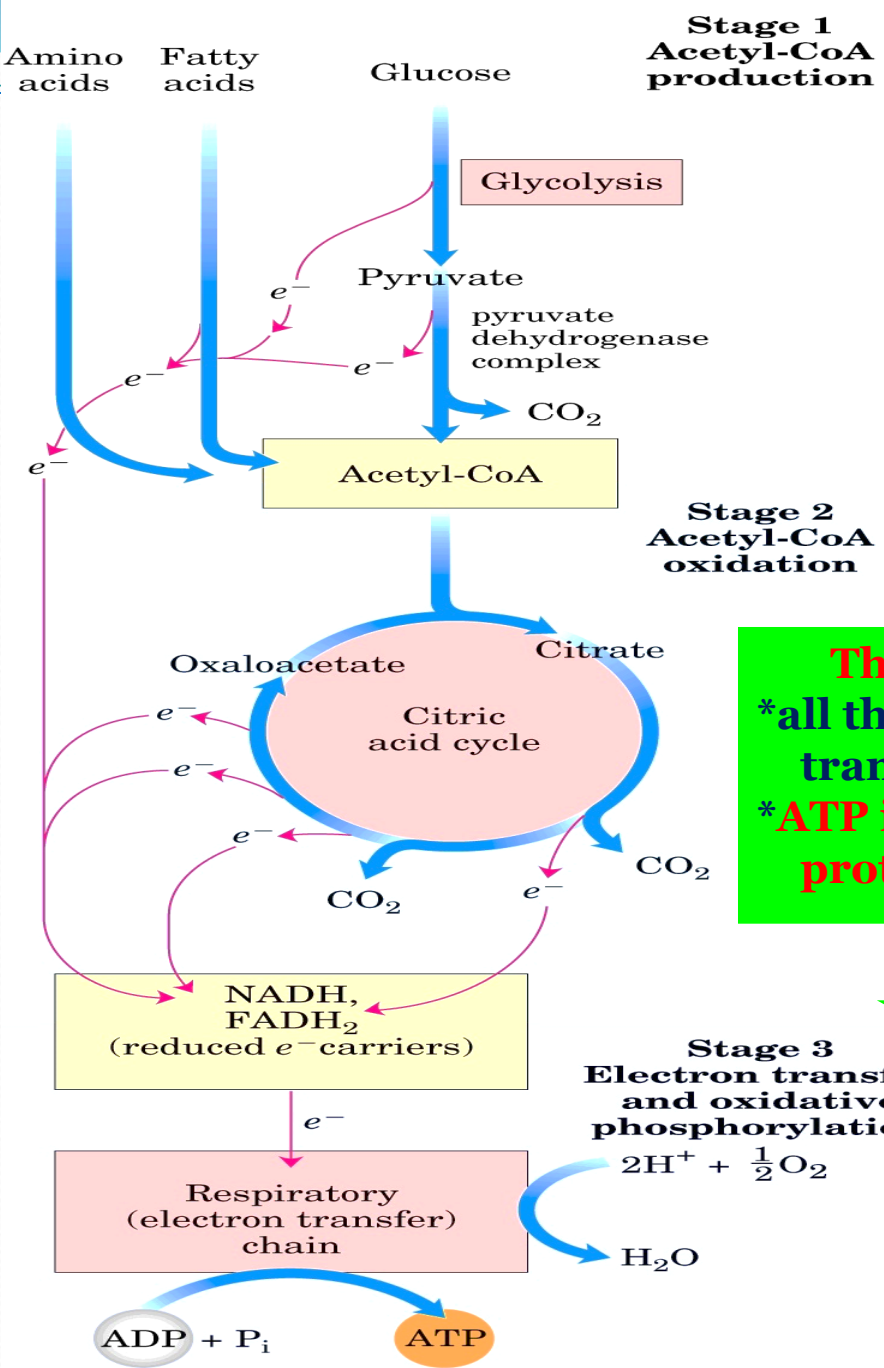
Components of ETC are arranged in order of increasing redox potential.

Electron pass on from electronegative NADH to electropositive O₂.

Electron transfer to O₂ is highly exergonic.

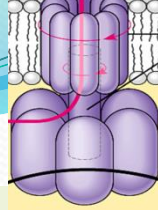
Called respiratory chain because of the reduction of O₂ from respiration into H₂O.

95% of oxygen consumed by humans is reduced to H₂O by cytochrome oxidase (300 ml H₂O/day) and called metabolic water.



The Main Event
 *all the electrons are transferred to O₂;
 *ATP is made using a proton gradient.

5 Correlation between Electron Transport system & oxidative phosphorylation

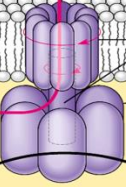


- **Electron Transport:** Electrons carried by reduced coenzymes are passed through a chain of proteins and coenzymes (in ETC) to drive the generation of a **proton gradient** across the **inner** mitochondrial membrane.
- **Oxidative Phosphorylation:** The proton gradient runs downhill to drive the synthesis of ATP.

In biologic systems:

- Cells use electron transport chain to transfer electrons **stepwise** from substrates to oxygen.
- Thus producing energy **gradually**.
- This process is **stepwise**, **efficient** and **controlled**.
- During hydrogen (H^+ and electron) transfer through different components of the redox chain, energy is released **gradually** in **small** utilizable amounts instead of a **massive** energy production in the form of **heat**, which if happens may **destroy** the living cells.

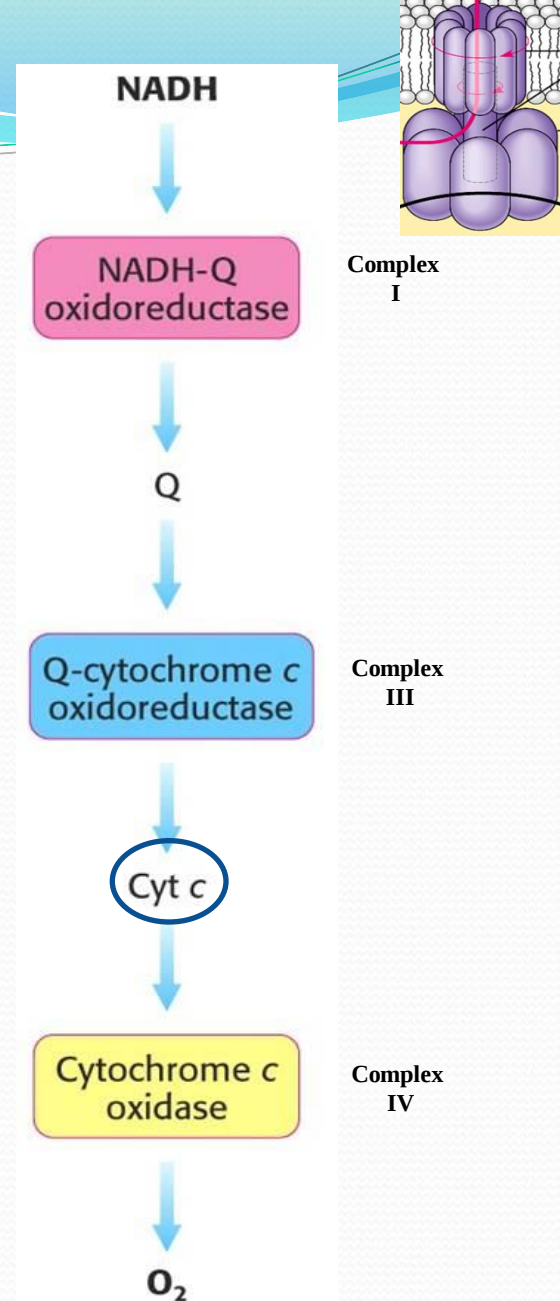
ETC Complexes



- **Four** protein complexes (**I to IV**) in the **inner** mitochondrial membrane and **one ATP synthase complex**.
- A **lipid** soluble coenzyme (**UQ, CoQ**) and a **water** soluble protein (**cytc**) shuttle between protein complexes.
- Electrons generally **fall or flow** in energy through the chain - from complexes I and II to complex III → IV (**$RH_2 \rightarrow H^+ e^- \rightarrow O_2$**).
- **Complex I = NADH-CoQ10 oxidoreductase** (Electron transfer from NADH to CoQ10) = **4H+ pumped**
 - This complex accept H^+ and Hydride ion from reduced NAD.
- **Complex II = succinate dehydrogenase (succinate CoQ10 oxidoreductase)**
 - This complex accept H^+ and Hydride ion from reduced FAD and **no H+ pumped**.
 - **Co Q: Lipid** soluble Ubiquinone called Coenzyme Q that accept H atoms from complex I and II to **transfer** it into complex III.

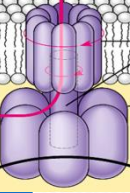
ETC Complexes

- **Complex III = CoQ10-Cytochrome c oxidoreductase** CoQ10 (contains cytochromes, **b** and **c**) passes electrons to Cyt c (and pumps H^+) in a unique redox cycle known as the Q cycle. **$4H^+$ pumped.**
- **Cyt c:** is a water-soluble electron carrier, transfer electrons from complex III to complex IV.
- **Complex IV = Cytochrome oxidases** ($a+a_3$ and **copper** center). Electrons from cyt c are used in a four-electron reduction of O_2 to produce $2H_2O$. O_2 is the final electron acceptor. **$2H^+$ pumped.**

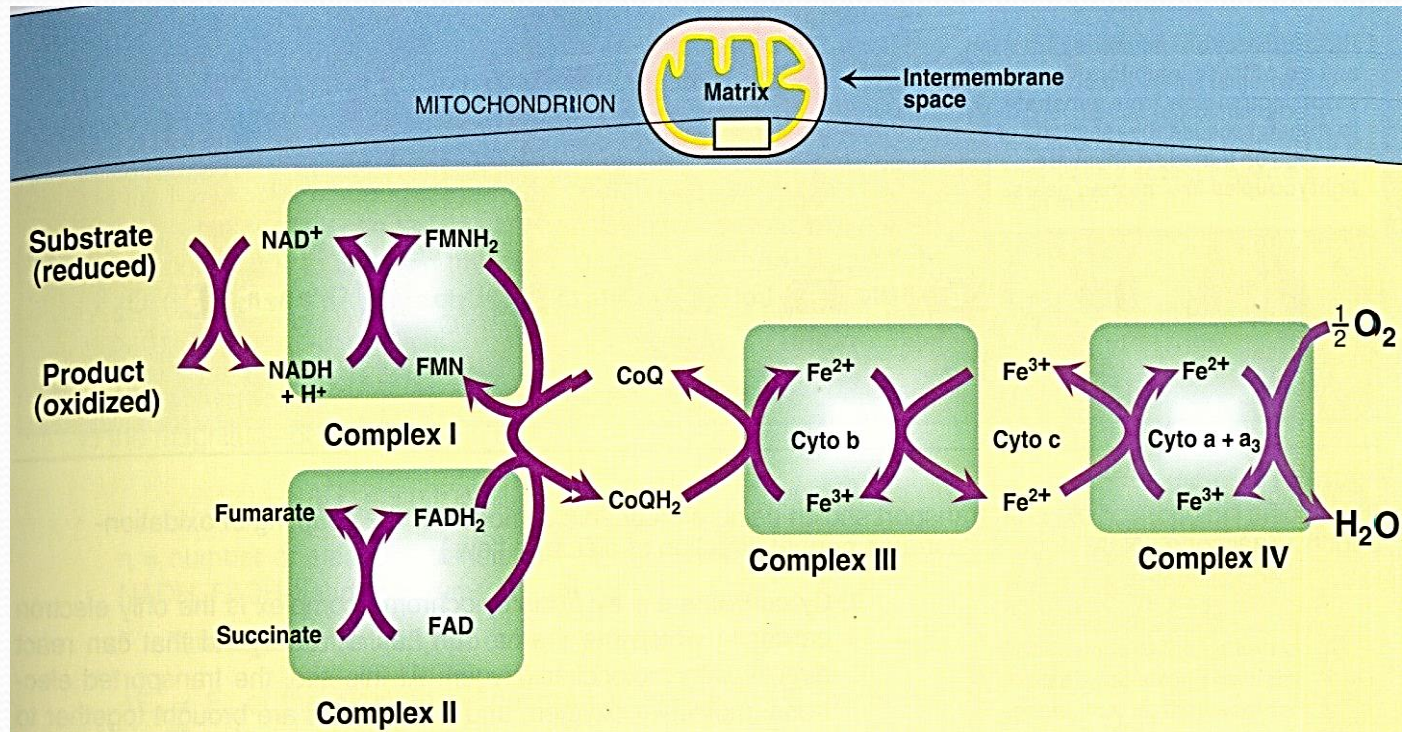


Mobile Electron Carriers

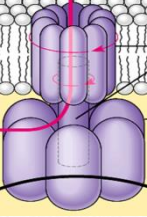
CoQ and Cyt c



CoQ (Ubiquinone)	Cyt C
Lipid soluble mobile electron carrier	Water soluble mobile electron carrier
Organic molecule (not a protein).	Metallo-protein.
Carry electron from Complex I or II to complex III	Carry electron from Complex III to complex IV

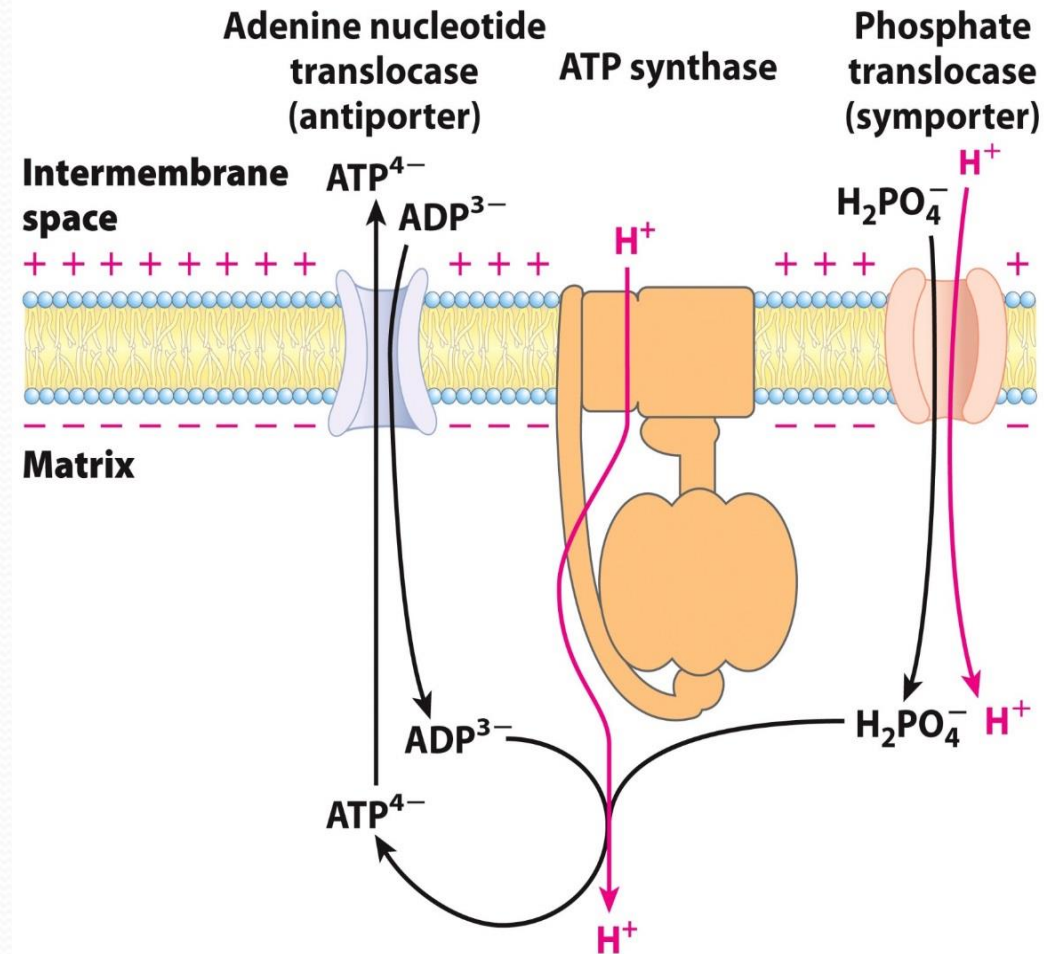


Complex V = ATP Synthase

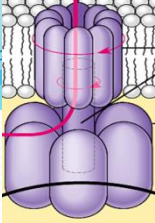


It is H^+ channel responsible for the Coupling of the energy from e^- Transport and H^+ flow with **oxidative phosphorylation** to produce energy as **ATP**.

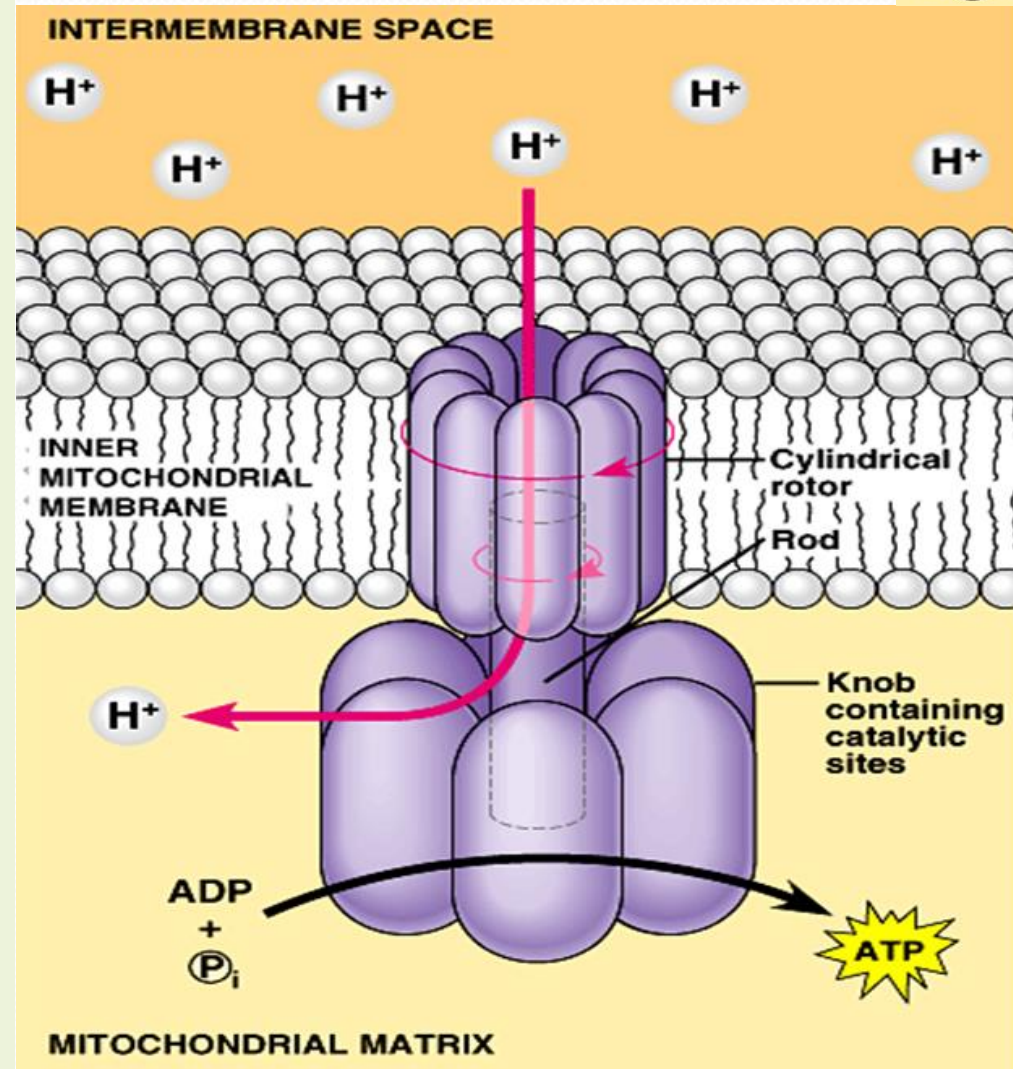
The **enzyme** use the **proton gradient** across the inner membrane to drive the synthesis of ATP



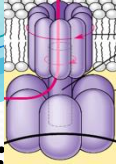
Mitchell's hypothesis (chemiosmosis model)



- Complex I, III and IV act as **proton pumps**.
- The translocation of protons H^+ from the mitochondrial matrix into the inter-mitochondrial space is called (**proton pumping**)
- H^+ pumping & electron transport results in an **electrochemical gradient**



Chemiosmotic Hypothesis

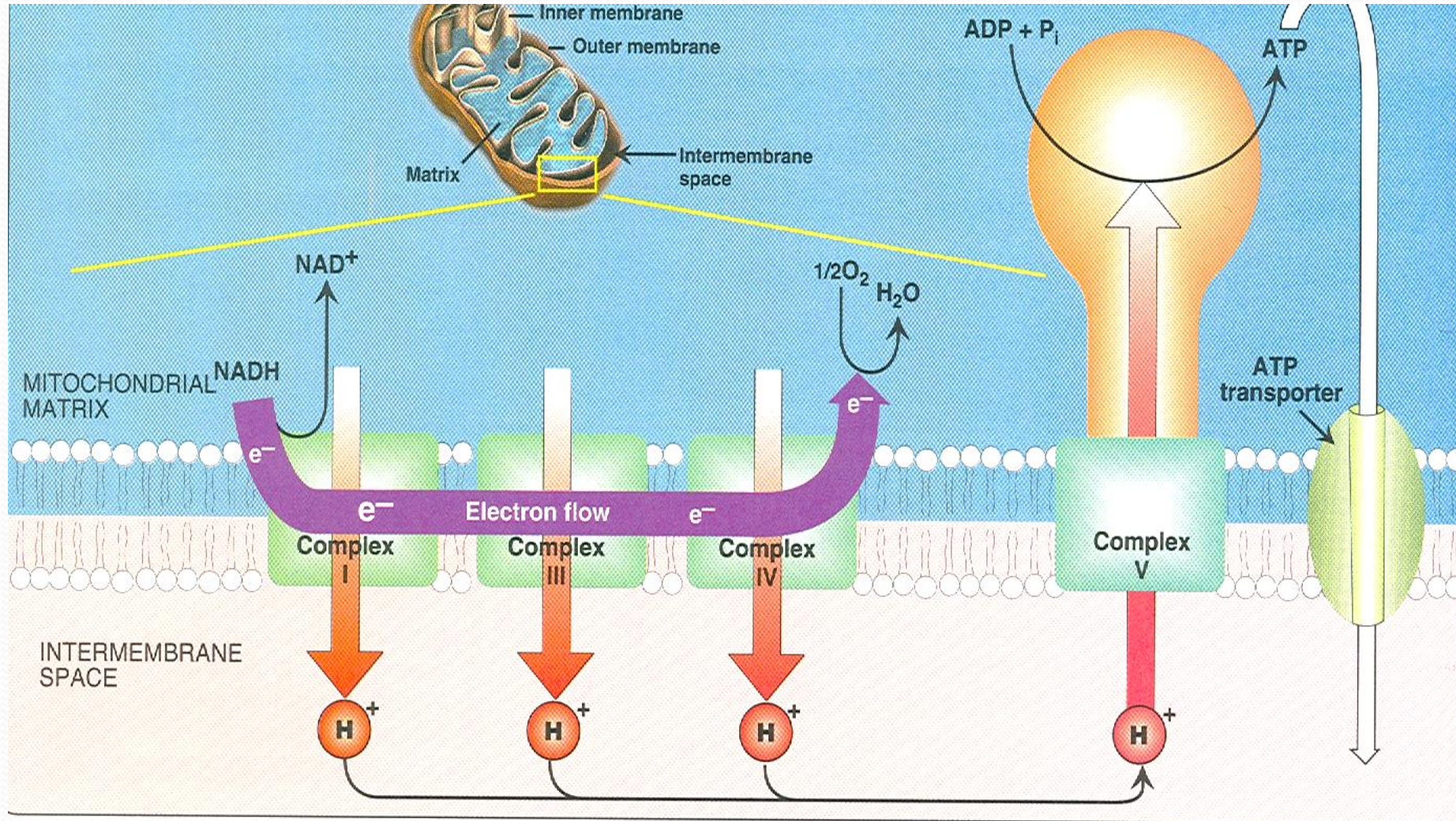
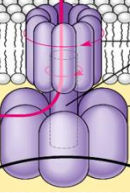


- **Proton motive force**: energy released by flow of H^+ down its gradient is used for ATP synthesis.
- The energy obtained from **electron transport** is coupled to the **proton motive force** in what's called **Chemiosmosis**.
- Mitchell proposed that a proton gradient across the inner membrane could be used to drive ATP synthesis.
- More +ve on the outside of the membrane than on the inside → **Electrochemical gradient**.
- Energy generated by **Electrochemical gradient** is sufficient to drive ATP synthesis i.e. **couples oxidation to phosphorylation**.

Findings to support chemiosmosis model

1. Addition of protons (acid) to the external medium of the mitochondria stimulates ATP production.
2. Oxidative phosphorylation does **not** occur in case of solubilising mitochondrial membranes.
3. **Uncouplers**.

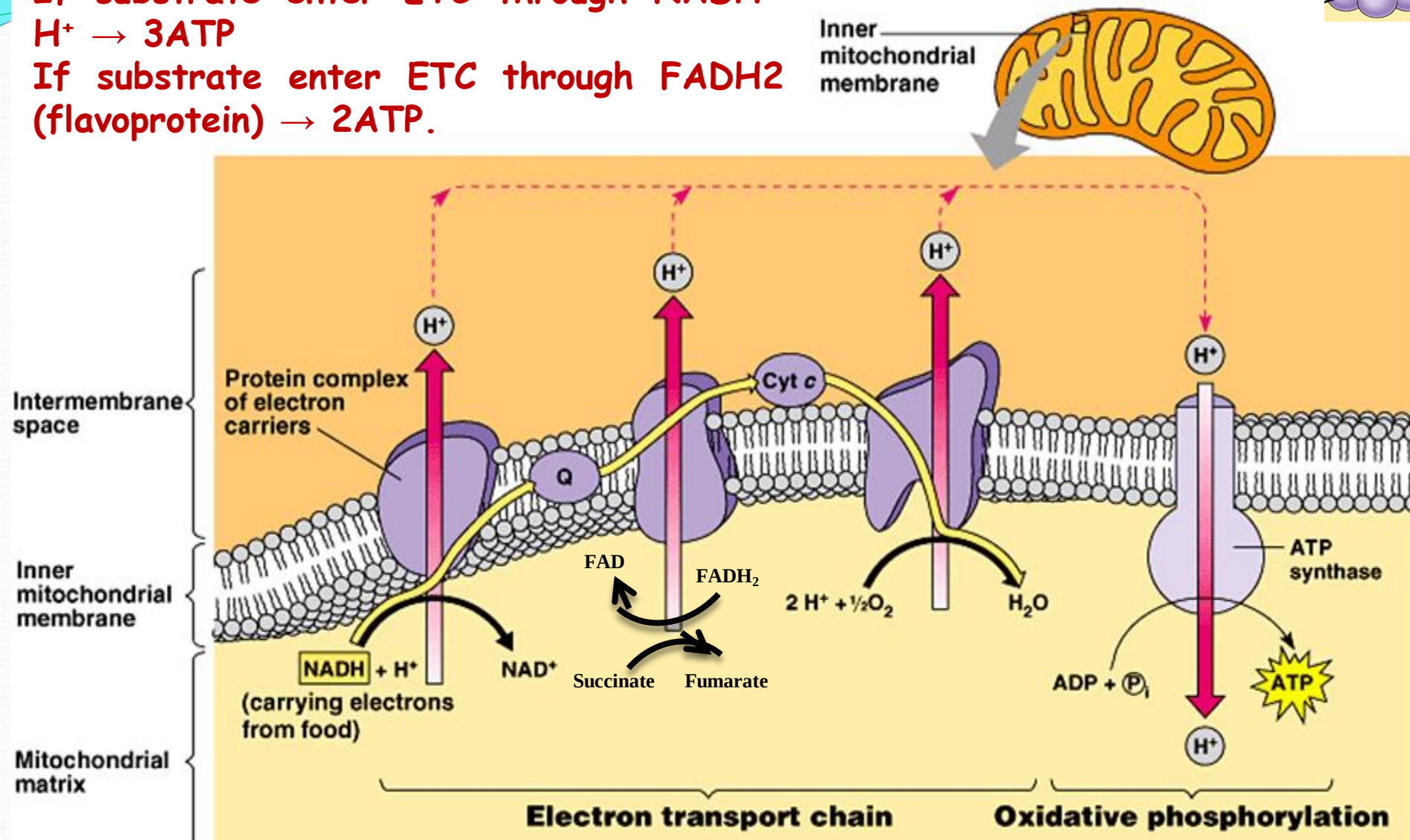
Chemiosmotic Hypothesis

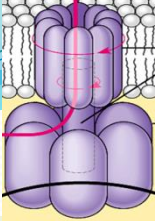


Summary of ETC and oxidative phosphorylation

If substrate enter ETC through NADH+
 $H^+ \rightarrow 3ATP$

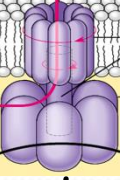
If substrate enter ETC through FADH₂
 (flavoprotein) $\rightarrow 2ATP$.





- There is **no storage** form for ATP.
 - So **all** ATP formed is only present to **cover** the needs of the cell at the **moment** as a source of Energy.
 - This is why there should be a **controlled** way for the production of ATP under electron transport chain (ETC).
1. **Availability of ADP** (ADP/ATP transporter is a rate limiting step in ETC).
 2. **Availability of electrons:**
 - NADH/NAD ratio or FADH₂/FAD ratio.
 3. **Availability of O₂.**

Uncouplers



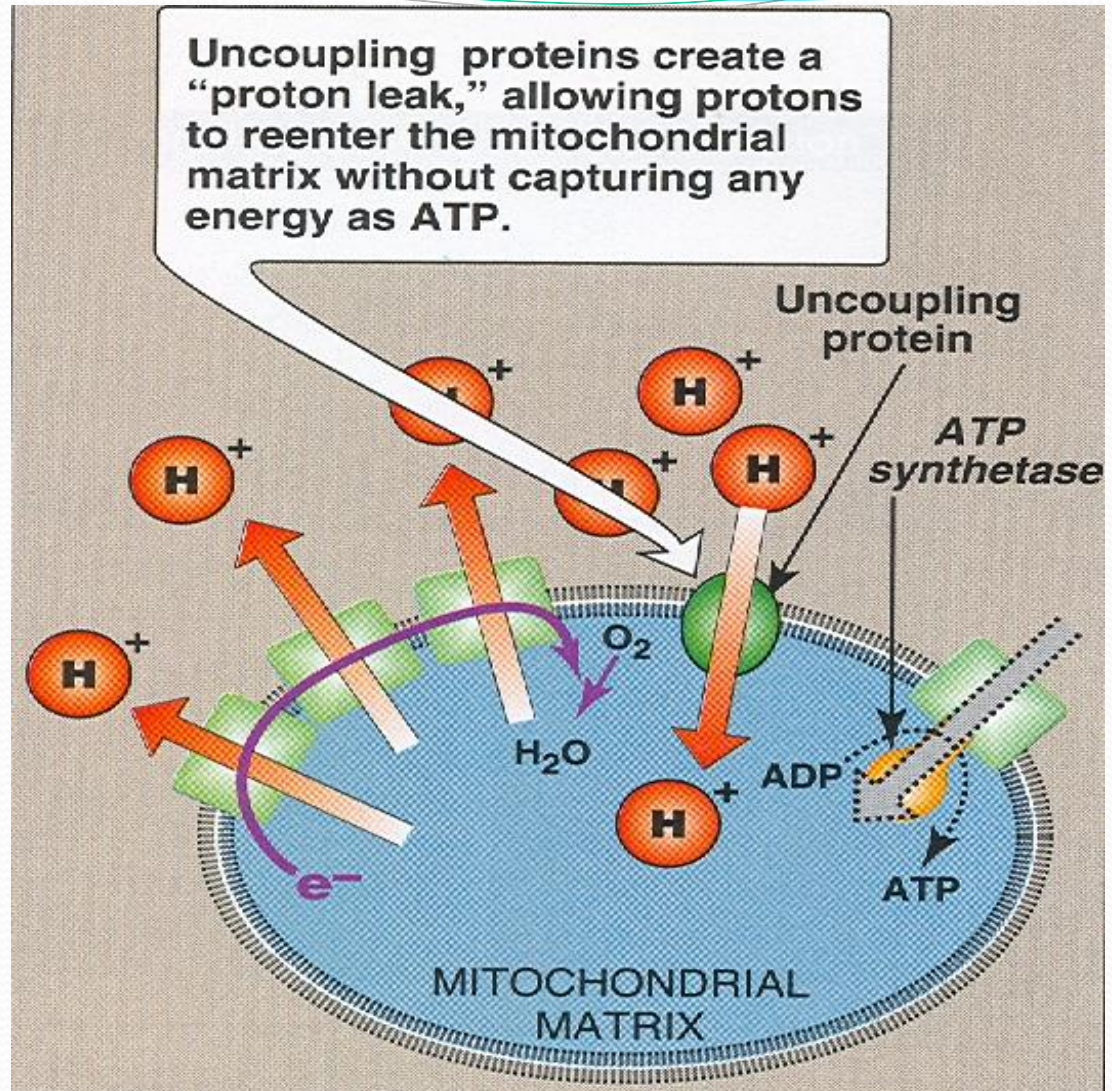
- These compounds **abolished** the **coupling** between oxidation and phosphorylation through **increasing** the **permeability** of the intra-membrane space → Failure of the electrochemical gradient formation → ATP formation **stops** while oxidation proceeds → Energy is released as **heat** rather than ATP.

Physiological Uncoupling

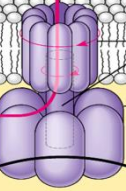
- An **uncoupling protein** (**thermogenin**) is produced physiologically in **brown adipose tissue** of **newborn** mammals including human → this protein is in inner mitochondrial membrane → This protein is **H⁺ carrier** → blocks development of a H⁺ electrochemical gradient → energy of respiration is dissipated as **heat** rather than ATP.

Toxic Uncoupling (DNP)

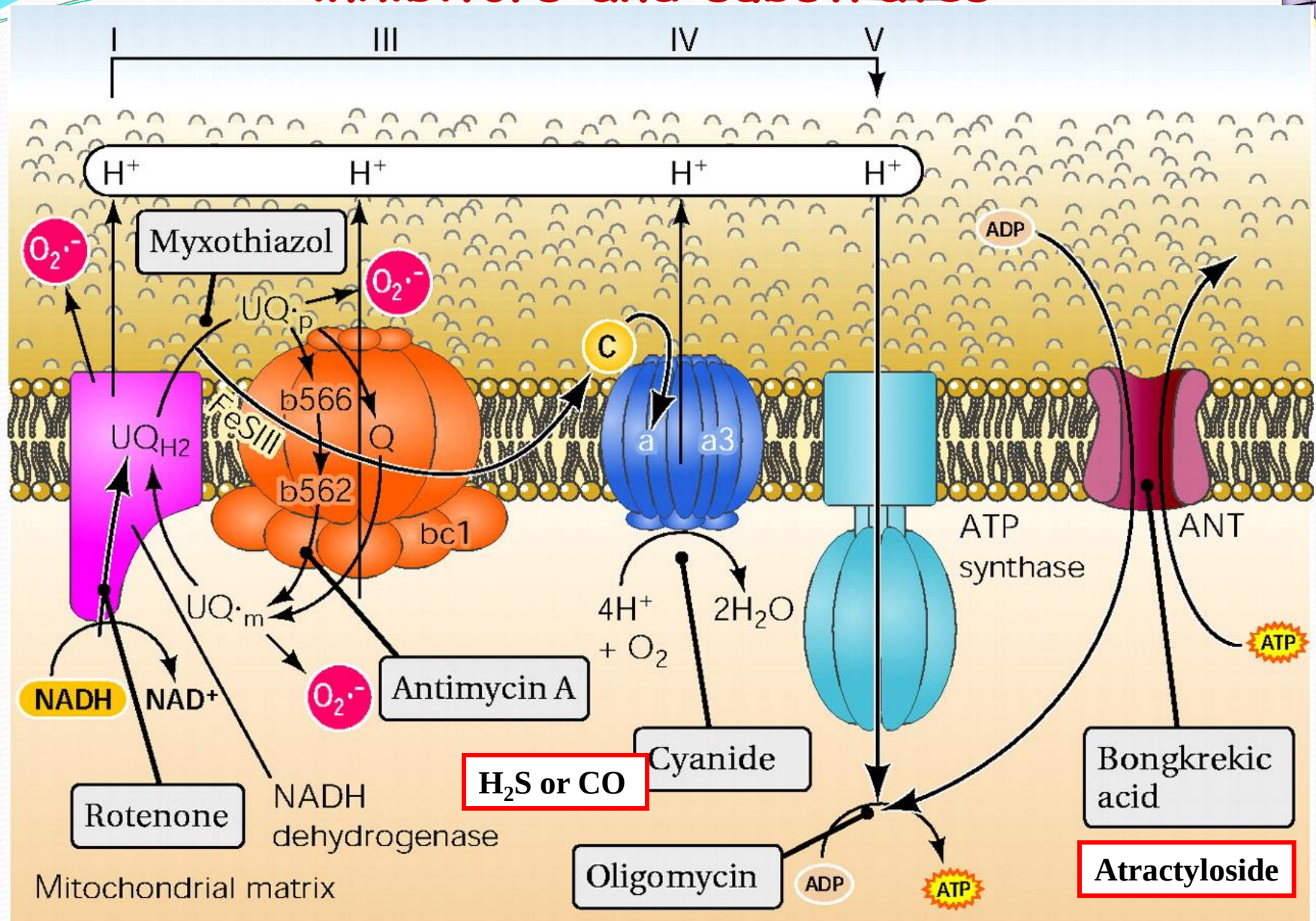
- N.B: DNP, thyroid hormones (**hyper-thyroidism**), high doses of **aspirin** and **arsenate** are toxic uncouples → feeling of increased body temperature (hotness) and **weight lose**.



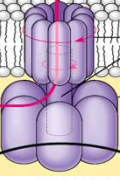
Transport of H^+ across mitochondrial membrane by 2,4-dinitrophenol.



Electron transport chain inhibitors and substrates



18 Inhibitors and uncouplers of oxidative phosphorylation



Inhibitors:

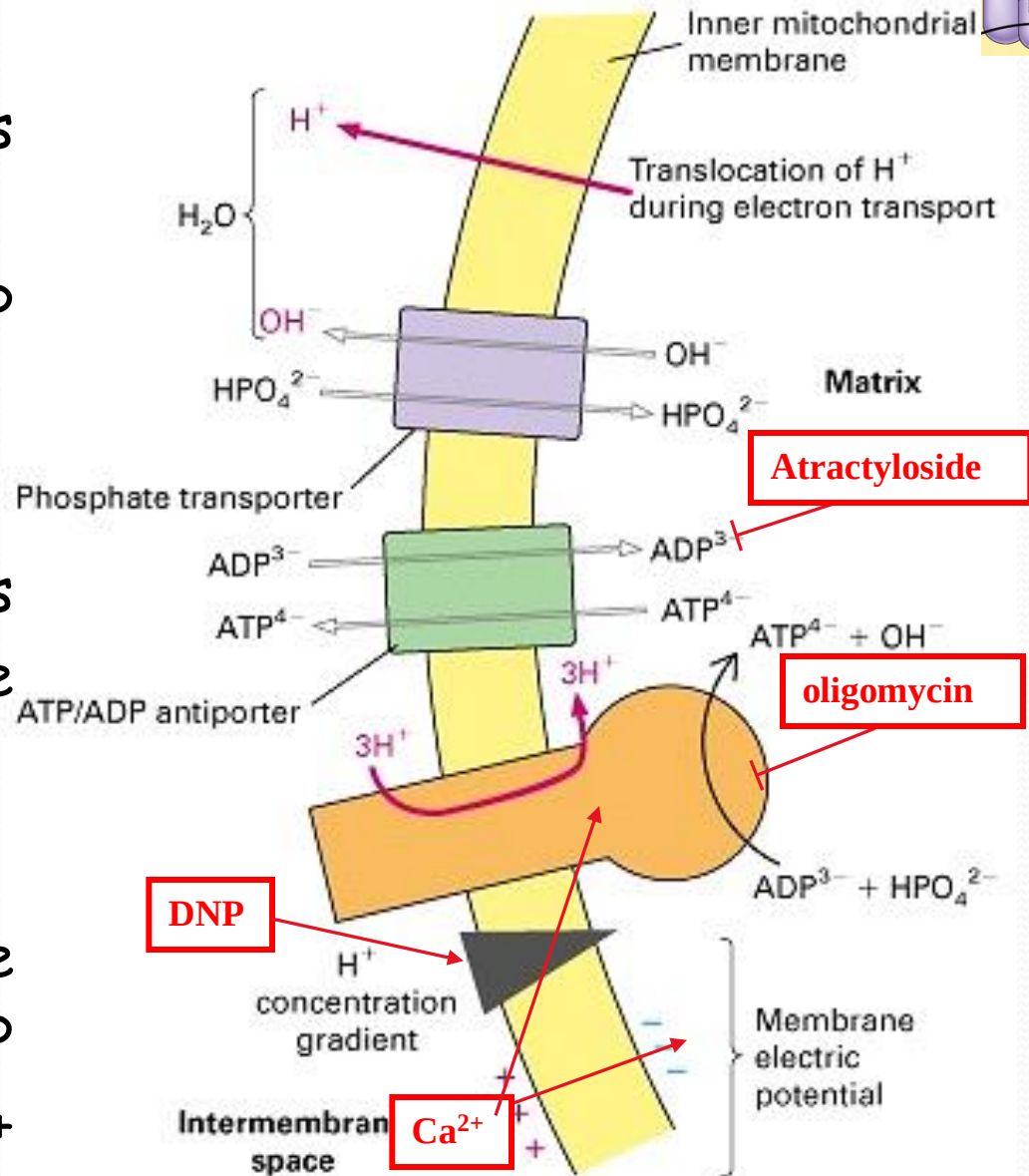
- **Atractyloside**: inhibits ADP/ATP antiporter
- **Oligomycin**: inhibits ATP synthase (**Uncoupler**).

Toxic Uncoupler:

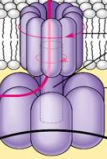
- **DNP** shuttles H^+ across inner membrane, remove potential gradient

$CaCl_2$

- It **stimulates** oxidative phosphorylation and ATP production (++ F0-F1, ++ dehydrogenases).

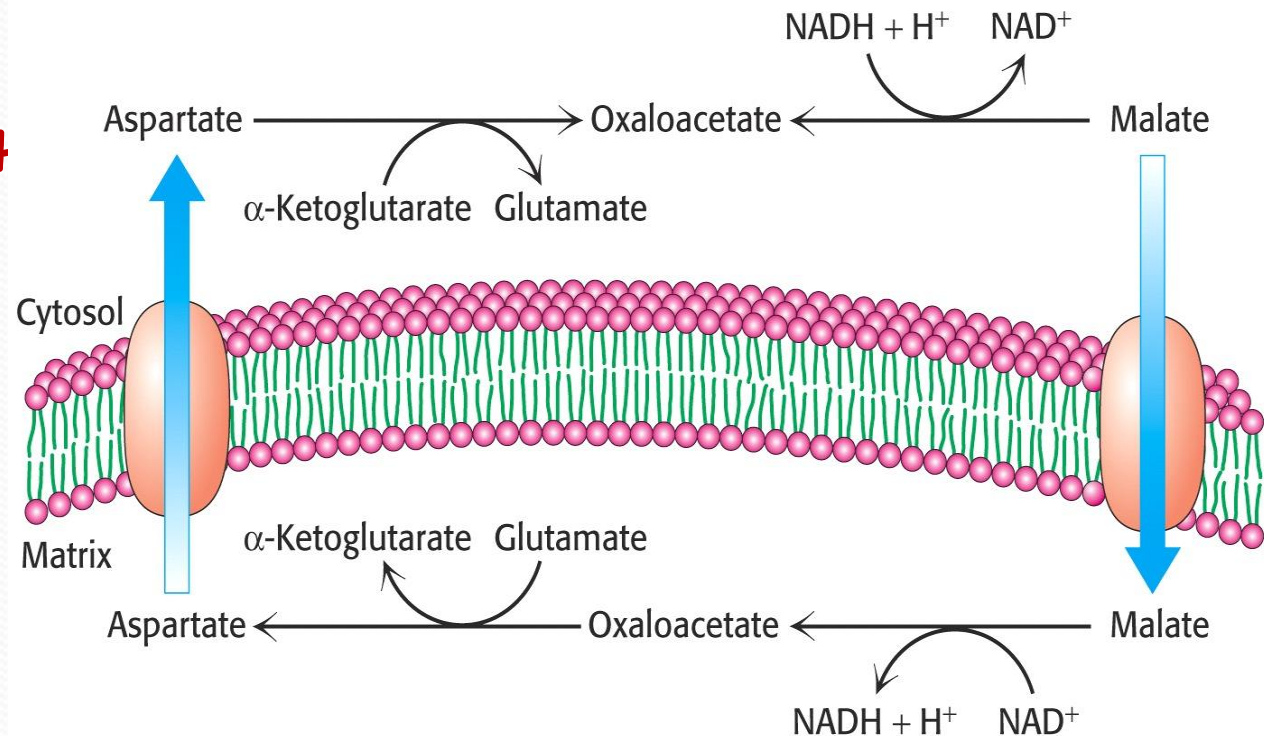


Oxidation of Extra-Mitochondrial NAD



- Some NADH molecules are reduced in the **cytosol** and must be **transported** into the **mitochondria** for electrons to enter the electron transport pathway.
- Two different "**shuttles**" are commonly encountered:
 - Glycerol 3-phosphate shuttle** (transfers electrons to **FADH₂**).
 - Malate-aspartate shuttle** (transfers electrons to **NADH**).

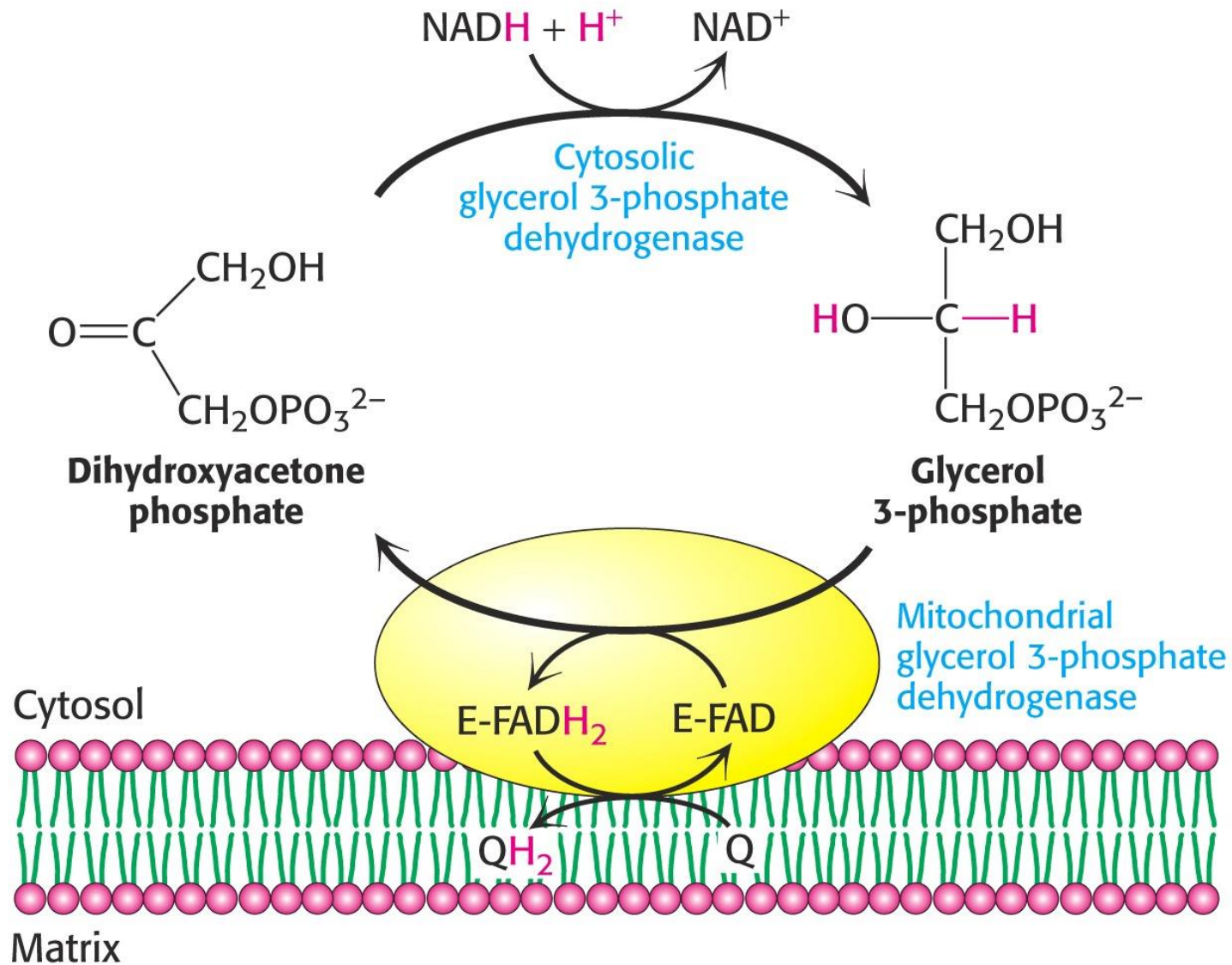
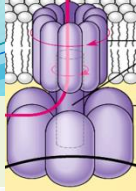
Malate- Aspartate shuttle: (NADH→NAD⁺)



In eukaryotes → 38 ATP.

20 Glycerol 3-phosphate shuttle: ($\text{NADH} \rightarrow \text{FADH}_2$)

In plants, fungi and some animals $\rightarrow$ 36 ATP.



THANK
YOU

A decorative illustration of a branch with red and pink leaves and small green berries, framing the text 'THANK YOU'. The leaves are in various shades of red and pink, with some having white veins. The berries are small and green, clustered on thin stems. The text 'THANK YOU' is written in a black, serif font, with 'THANK' on the top line and 'YOU' on the bottom line. The words are slightly tilted and framed by the foliage.